THE WALTER-BERTINI THEOREM VIA JET SCHEMES

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Contents

1. Preliminaries 2
2. Review of sheaves of principal parts 5
3. The differential of a family of jets of sections 8
4. Proof of Theorem 0.1 12

References 15

In this note we illustrate the method of the paper [Bra19] by using it to establish the following Bertini-type result, which was first proved by Walter [Wal94]:

{thm:intro}

Theorem 0.1. *Let X be a smooth algebraic scheme over an infinite field. Let E be a locally free sheaf on X. Let $Z \subseteq \mathbf{V}(E)$ be a subscheme of the vector bundle associated to E. Suppose that Z is smooth over X. Let $W \subseteq \Gamma(X, E)$ be a finite-dimensional linear subspace that separates jets of order 1 (see Definition 1.9). Then a general section $s \in W$ is transverse to Z.*

When the base field is algebraically closed, to say that W separates jets of order 1 is to say that that the natural map $W \to E/\mathfrak{m}_x^2 E$ is surjective for all closed points $x \in X$. If the locally free sheaf E is of rank 1, this may be reprased as "the linear system $(E, \mathbf{P}(W^\vee))$ separates tangent vectors".

The conclusion of Theorem 0.1 says that the inverse image $s^{-1}Z \subseteq X$ is either empty or smooth and of codimension in X equal to the codimension of Z in $\mathbf{V}(E)$, see Remark 4.2. Taking $Z \subseteq \mathbf{V}(E)$ to be the zero section, we obtain the following corollary.

{cor:intro-vanishing}

Corollary 0.2. *Let X be a smooth algebraic scheme over an infinite field. Let E be a locally free sheaf on X. Let $W \subseteq \Gamma(X, E)$ be a finite-dimensional linear subspace that separates jets of order 1. Then the vanishing locus of a general section $s \in W$ is either empty or smooth of codimension in X equal to the rank of E.*

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Corollary 0.2 generalizes the result of Spreafico mentioned above. Indeed, if X is projective and $E(-1)$ is globally generated, then $\Gamma(X, E)$ separates jets of order 1.

Another application of Theorem 0.1 is to degeneracy loci. By definition, the *ith degeneracy locus* of a map $\alpha : E \to F$ between locally free sheaves on a scheme X is the the closed subset

$$\{x \in X \ : \ \alpha(x) : E(x) \to F(x) \text{ has rank at most } \min(e, f) - i\} \subseteq X$$

equipped with its natural determinantal scheme structure (see Definition 1.4). We denote this subscheme of X by $\Sigma^i(\alpha)$.

{cor:intro-deg}

Corollary 0.3. *Let X be a smooth algebraic scheme over an infinite field. Let E and F be locally free sheaves on X of respective ranks e and f. Let $W \subseteq \Gamma(X, \mathcal{H}om_X(E, F))$ be a finite-dimensional linear subspace that separates jets of order 1. Let i be a nonnegative integer. If $\alpha \in W$ is a general section, then the constant-rank locus $\Sigma^i(\alpha) \setminus \Sigma^{i+1}(\alpha)$ is either empty or smooth and of codimension $i(|e - f| + i)$ in X.*

Proof. Follows from the combination of Theorem 0.1, Proposition 1.7, and Remark 1.6. □

Corollary 1.8 generalizes Corollary 0.2, since the vanishing locus of a section $s \in \Gamma(X, E)$ equals the first degeneracy locus of the map $\mathcal{O}_X \to E$ that sends $f \mapsto f \cdot s$.

1. Preliminaries

1.1. **Vector bundles.** Our convention for the correspondence between locally free sheaves and vector bundles is different from Grothendieck's [EGA, Définition II.1.7.8]. Let X be a scheme. Let E be a locally free $\mathcal{O}_X$-module of finite rank. To us, the *vector bundle corresponding to E* is the X-scheme

$$\mathbf{V}(E) := \operatorname{Spec}_X \operatorname{Sym}(E^\vee).$$

With this convention, there is a natural isomorphism between E and the sheaf of sections of the projection $\pi : \mathbf{V}(E) \to X$. Indeed, given a morphism of schemes $t : T \to X$, there are natural bijections

$$\begin{aligned}\operatorname{Hom}_X(T, \mathbf{V}(E)) &= \operatorname{Hom}_{\mathcal{O}_X\text{-alg.}}(\operatorname{Sym}(E^\vee), t_*\mathcal{O}_T)\\ &= \operatorname{Hom}_{\mathcal{O}_X}(E^\vee, t_*\mathcal{O}_T)\\ &= \operatorname{Hom}_{\mathcal{O}_T}(t^*E^\vee, \mathcal{O}_T)\\ &= \Gamma(T, t^*E)\end{aligned}$$

by the universal mapping properties of the relative spectrum and of the symmetric algebra, and the adjunction between pullback and pushforward.

{def:taut-sect}

Definition 1.1. The *tautological section* $\tau \in \Gamma(\mathbf{V}(E), E_{\mathbf{V}(E)})$ is the section corresponding to the identity morphism of $\mathbf{V}(E)$.

The map

$$\mathrm{Hom}_X(T, \mathbf{V}(E)) \to \Gamma(T, t^*E)$$

given by pullback of the tautological section coincides with the natural bijection described above.

The components of the tautological section with respect a trivialization of E are "coordinates along the fibers" of $\pi : \mathbf{V}(E) \to X$. More precisely:

{vb-trivial}

Example 1.2. Suppose that the $\mathcal{O}_X$-module E is free with basis $\{v_1, \dots, v_e\} \subseteq \Gamma(X, E)$. Let $\mathbf{A}^e$ be the affine space over $\operatorname{Spec} \mathbf{Z}$ with coordinates $t_1, \dots, t_e$. Then there exists a unique isomorphism of X-schemes $\mathbf{V}(E) \cong X \times \mathbf{A}^e$ with respect to which

$$\tau = t_1 \cdot \pi^* v_1 + \cdots + t_e \cdot \pi^* v_e.$$

1.2. **A Bertini-type statement.** The conclusion of our main result asserts the transversality of the intersection between a section of a vector bundle and a subvariety of that vector bundle. This assertion has two components: first, that the intersection is smooth and, second, that it is of the expected dimension.

The next result can be used to control the dimension of such intersections in arbitrary characteristic, and their smoothness in characteristic zero. It is a slight variant of [Ful98, Lemma B.9.1], which is in turn a variant of a result of Atiyah-Serre [Ati57, Part 1, §5].

{serre-bertini}

Proposition 1.3. *Let X be a scheme of finite type over an infinite field k. Let E be a locally free $\mathcal{O}_X$-module. Let $\Sigma \subseteq \mathbf{V}(E)$ be a locally closed subscheme of the vector bundle associated to E. Suppose that X and Σ are pure-dimensional, and let c denote the codimension of Σ in $\mathbf{V}(E)$. Let $W \subseteq \Gamma(X, E)$ be a finite-dimensional k-linear subspace that generates E. Let $s \in W$ be a general section. Then $s^{-1}\Sigma \subseteq X$ is either empty or of pure codimension c. Furthermore, if Σ is smooth over k and k has characteristic zero, then $s^{-1}\Sigma$ is smooth over k.*

Proposition 1.3 has a simple proof, but already generalizes the characteristic-zero case of main result of this paper. Indeed, Theorem 0.1 requires that the X be smooth and Σ be smooth over X, while Proposition 1.3 yields transversality in characteristic zero under the weaker assumption that Σ is smooth.

Proof of Proposition 1.3. We repeat the argument of [Ful98, Lemma B.9.1]. Let n denote the dimension of X. Let d denote the dimension of W as a vector space over k. By assumption, the natural map $\alpha : W \otimes_k \mathcal{O}_X \to E$ is surjective. This implies that the morphism of vector bundles $\tilde{\alpha} : W \times_k X \to \mathbf{V}(E)$ induced by α is smooth and surjective. It follows that the inverse image $\tilde{\alpha}^{-1}\Sigma$ has pure dimension $d + n - c$, and is smooth over k if Σ is. If $s \in W$, then $s^{-1}\Sigma$ is the pullback of $\tilde{\alpha}^{-1}\Sigma$ along the morphism $\operatorname{Spec} k \to W$

corresponding to s.

$$\begin{tikzcd}
s^{-1}\Sigma \arrow[r] \arrow[d, hook] & \tilde{\alpha}^{-1}\Sigma \arrow[r] \arrow[d, hook] & \Sigma \arrow[d, hook] \\
X \arrow[r] \arrow[d] & W \times_k X \arrow[r, "\tilde{\alpha}"] \arrow[d] & \mathbf{V}(E) \\
\operatorname{Spec} k \arrow[r, "s"] & W &
\end{tikzcd}$$

The proposition is clear if the first projection $\tilde{\alpha}^{-1}\Sigma \to W$ is not dominant. Suppose that it is. Then the result follows from well-known theorems on generic flatness or, in characteristic zero, generic smoothness applied to this projection. $\square$

1.3. **Degeneracy loci.** Let X be a scheme. Let E and F be locally free $\mathcal{O}_X$-modules of ranks e and f. Let $m = \min(e, f)$. Let i be a nonnegative integer.

{degeneracy-locus-def}

Definition 1.4. The *ith degeneracy locus* of an $\mathcal{O}_X$-linear map $\alpha : E \to F$ is defined to be the subscheme $\Sigma^i(\alpha) \subseteq X$ where exterior power

$$\wedge^{m-i+1}\alpha : \wedge^{m-i+1}E \to \wedge^{m-i+1}F$$

vanishes if $i \leq m+1$, and the empty scheme otherwise.

{deg-loci-functor}

Remark 1.5.
(1) Locally, the ideal of $\Sigma^i(\alpha) \subseteq X$ is generated by all minors of size $m - i + 1$ of a matrix representation for α.
(2) A point $x \in X$ lies in $\Sigma^i(\alpha)$ if, and only if, the $k(x)$-linear map $\alpha(x)$ has rank at most $m - i$.
(3) By the Laplace expansion of the determinant, we have closed immersions

$$\varnothing = \Sigma^{m+1}(\alpha) \subseteq \Sigma^m(\alpha) \subseteq \cdots \subseteq \Sigma^0(\alpha) = X.$$

(4) If $t : T \to X$ be a morphism of schemes, then

$$t^{-1}\Sigma^i(\alpha) = \Sigma^i(t^*\alpha)$$

as closed subschemes of T.

Let $\pi : V \to X$ be the vector bundle corresponding to the locally free $\mathcal{O}_X$-modules $\mathcal{H}om_X(E, F)$. In symbols,

$$V = \mathbf{V}(\mathcal{H}om_X(E, F)).$$

Let $h : E_V \to F_V$ be the tautological map, see Definition 1.1.

{rmk:univ-deg-loci}

Remark 1.6. The degeneracy loci of h are universal in the sense that if $\alpha : E \to F$ is an $\mathcal{O}_X$-linear map, then $\Sigma^i(\alpha) = \alpha^{-1}\Sigma^i(h)$, where on the right we view α as a morphism $X \to V$.

The following result is well known:

{univ-deg-loci}

Proposition 1.7. *Suppose that $i \leq m$. The constant-rank locus $\Sigma^i(h) \setminus \Sigma^{i+1}(h) \subseteq V$ is smooth over X and of pure relative codimension $i(|e-f|+i)$ in V over X.*

We conclude this subsection with the following Bertini-type result for degeneracy loci:

{cor:intro-deg}

Corollary 1.8. *Let X be a smooth algebraic scheme over an infinite field k. Let E and F be locally free $\mathcal{O}_X$-modules of ranks e and f. Let $m := \min(e, f)$. Let $W \subseteq \Gamma(X, \mathcal{H}om_X(E, F))$ be a finite-dimensional linear subspace. Suppose that*

- *W generates $\mathcal{H}om(E, F)$ if k has characteristic zero; and*
- *W separates 1-jets if k has positive characteristic.*

Let $\alpha \in W$ is a general section. Then the constant-rank locus $\Sigma^i(\alpha) \setminus \Sigma^{i+1}(\alpha)$ is either empty or smooth and of relative dimension $i(|e-f|+i)$ in X.

Proof. This follows from the combination of Theorem 0.1 (or Proposition **??** below), Propositions 1.3 and 1.7, and Remark 1.6. □

1.4. **Separation of jets.** Let X be a scheme over a field k. Let E be an $\mathcal{O}_X$-module. Let $W \subseteq \Gamma(X, E)$ be a linear subspace. Let m be a nonnegative integer.

{def:separates-jets}

Definition 1.9. If k is algebraically closed, we say that W *separates jets of order m* if the natural map $W \to E/\mathfrak{m}_x^{m+1}E$ is surjective for all closed points $x \in X$. In general, we say that W *separates jets of order m* if there exists an algebraic closure $k \subseteq \bar{k}$ such that $W \otimes_k \bar{k} \subseteq \Gamma(X \times_k \bar{k}, E_{X \times_k \bar{k}})$ separates jets of order m.

Proposition 2.1 below shows that if W separates jets of order m, then $W \otimes_k K \subseteq \Gamma(X \times_k K, E_{X \times_k K})$ separates jets of order m for *every* field extension $k \subseteq K$. That proposition also shows that Definition 2.1 may be reformulated in terms of sheaves of principal parts, a fact that we use in the proof of Theorem 0.1.

2. Review of sheaves of principal parts

The theory of Grothendieck's sheaves of principal parts is developed in [EGA, Chapter IV]. In this section we collect basic facts about these sheaves and prove a result concerning the notion of separation of jets introduced above (Definition 1.9).

Let X be a scheme over a field k. Let E be an $\mathcal{O}_X$-module. Let m be a nonnegative integer. The *mth sheaf of principal parts* associated to E is a sheaf of $\mathcal{O}_X$-modules that we denote by $\mathcal{P}_X^m E$ (or by $\mathcal{P}_{X/k}^m E$ when we wish to emphasize the base field). The fiber of $\mathcal{P}_X^m E$ over a k-rational point $x \in X$ is canonically identified with the k-vector space $E/\mathfrak{m}_x^{m+1}E$. More precisely,

the sheaf $\mathcal{P}^m_X E$ is equipped with a k-linear sheaf morphism $d^m_E : E \to \mathcal{P}^m_X E$ that induces a k-linear isomorphism

$$E/\mathfrak{m}_x^{m+1}E \xrightarrow{\sim} \mathcal{P}^m_X E \otimes k(x) \tag{2.1}$$ {eqn:fiber-pp}

for each k-rational point $x \in X$. The pair $(\mathcal{P}^m_X E, d^m_E)$ satisfies a *universal property*: $d^m_E : E \to \mathcal{P}^m_X E$ is the universal k-linear differential operator of order m from E to another $\mathcal{O}_X$-module, see [Stacks, Tags 09CK and 09CT].

Using this universal property, it is not hard to see that:

(1) The sheaf $\mathcal{P}^m_X E$ is generated as an $\mathcal{O}_X$-module by the image of the universal differential operator $d^m_E : E \to \mathcal{P}^m_X E$.

We write $\mathcal{P}^m_X$ and d^m_X instead of $\mathcal{P}^m_X(\mathcal{O}_X)$ and $d^m_{\mathcal{O}_X}$.

If $x \in X$ is a k-rational point, then the quotient $\mathcal{O}_X/\mathfrak{m}_x^{m+1}$ is not merely a k-vector space, but a k-algebra. Similarly, the quotient $E/\mathfrak{m}_x^{m+1}E$ is not merely a k-vector space, but a module over $\mathcal{O}_X/\mathfrak{m}_x^{m+1}$. These facts have counterparts for sheaves of principal parts:

(2) The $\mathcal{O}_X$-module $\mathcal{P}^m_X$ has a natural structure of $\mathcal{O}_X$-algebra. The multiplication map $\mathcal{P}^m_X \times \mathcal{P}^m_X \to \mathcal{P}^m_X$ is the unique $\mathcal{O}_X$-bilinear map such that
$$d^m_X u \cdot d^m_X v = d^m_X(uv)$$
for all sections $u, v \in \mathcal{O}_X$ defined over a common open subset of X.

(3) The $\mathcal{O}_X$-module $\mathcal{P}^m_X E$ has a natural structure of $\mathcal{P}^m_X$-module. The multiplication map $\mathcal{P}^m_X \times \mathcal{P}^m_X E \to \mathcal{P}^m_X E$ is the unique $\mathcal{O}_X$-bilinear map such that
$$d^m_X u \cdot d^m_E s = d^m_E(us)$$
for all sections $u \in \mathcal{O}_X$ and $s \in E$ defined over a common open subset of X.

We note that the universal derivation $d^m_X : \mathcal{O}_X \to \mathcal{P}^m_X$ is a map of k-algebras, but not of $\mathcal{O}_X$-algebras.

Sheaves of principal parts have many other properties that reflect familiar facts about their fibers. Below we'll need the following ones:

(4) There exists a unique isomorphism of $\mathcal{P}^m_X$-modules
$$\mathcal{P}^m_X \otimes_{d^m_X, \mathcal{O}_X} E \xrightarrow{\sim} \mathcal{P}^m_X E$$
that sends $\alpha \otimes s \mapsto \alpha \cdot d^m_E s$ for all local sections $\alpha \in \mathcal{P}^m_X$ and $s \in E$ defined over a common open subset of X.

(5) Let q be an integer such that $0 \leq q \leq m$. There exists a unique $\mathcal{O}_X$-linear *truncation map*
$$\varepsilon_{m,q} : \mathcal{P}^m_X E \to \mathcal{P}^q_X E$$
such that $\varepsilon_{m,q} \circ d^m_X = d^q_X$. The map $d^0_E : E \to \mathcal{P}^0_X E$ is an $\mathcal{O}_X$-linear isomorphism, which we use to identify $\varepsilon_{m,0}$ with an $\mathcal{O}_X$-linear map $\mathcal{P}^m_X E \to E$.

(6) Suppose that $m \geq 1$. There exists a unique $\mathcal{O}_X$-linear map $\iota_m : \mathrm{Sym}^m \Omega_X \otimes E \to \mathcal{P}^m_X E$ such that

$$\iota_m(du_1 \cdots du_m \otimes s) = (d^m_X u_1 - u_1) \cdots (d^m_X u_m - u_m) \cdot d^m_E s$$

for all local sections $u_1, \ldots, u_m \in \mathcal{O}_X$ and $s \in E$ defined over a common open subset of X. The sequence of $\mathcal{O}_X$-modules

$$0 \longrightarrow \mathrm{Sym}^m \Omega_X \otimes_{\mathcal{O}_X} E \xrightarrow{\iota_m} \mathcal{P}^m_X E \xrightarrow{\varepsilon_{m,m-1}} \mathcal{P}^{m-1}_X E \longrightarrow 0$$

is right-exact in general, and is exact if either $m = 1$ or X is smooth over k and E is locally free.

(7) Let F be an $\mathcal{O}_X$-module. There exists an unique $\mathcal{P}^m_X$-linear map

$$\mathcal{P}^m_X \mathcal{H}om_{\mathcal{O}_X}(E, F) \to \mathcal{H}om_{\mathcal{P}^m_X}(\mathcal{P}^m_X E, \mathcal{P}^m_X F)$$

such that

$$(d^m_{\mathcal{H}om(E,F)} \alpha)(d^m_E s) = d^m_F(\alpha s)$$

for all local sections $\alpha \in \mathcal{H}om(E, F)$ and $s \in E$ defined over a common open subset of X. This map is an isomorphism when E and F are locally free of finite rank.

The following proposition shows that our notion of separation of jets (Definition 1.9) is stable under change of base field and may be rephased in terms of sheaves of principal parts. It will be used in our proof of Theorem 0.1.

{prop:sufficient-jets}

Proposition 2.1. *Let $W \subseteq \Gamma(X, E)$ be a linear subspace. The following are equivalent:*

(1) *$d^m_E(W)$ generates $\mathcal{P}^m_X E$ as an $\mathcal{O}_X$-module.*
(2) *$W \otimes_k K \subseteq \Gamma(X \times_k K, E_{X \times_k K})$ separates jets of order m for some field extension $k \subseteq K$.*
(3) *$W \otimes_k K \subseteq \Gamma(X \times_k K, E_{X \times_k K})$ separates jets of order m for every field extension $k \subseteq K$.*

Proof. To emphasize the base field, in this proof we will write $\mathcal{P}^m_{X/k} E$ instead of $\mathcal{P}^m_X E$.

If k is algebraically closed, then $d^m_E(W)$ generates $\mathcal{P}^m_{X/k} E$ as an $\mathcal{O}_X$-module if, and only if, $W \subseteq \Gamma(X, E)$ separates jets of order m; this equivalence follows immediately from the canonical isomorphism $E/\mathfrak{m}^{m+1}_x E = \mathcal{P}^m_{X/k} E \otimes k(x)$ that exists for each closed point $x \in X$.

Let $k \subseteq K$ be a field extension. Let $X_K := X \times_k K$ and let E_K denote the pullback of E along the first projection $u : X_K \to X$. By the preceding paragraph, it suffices to prove that $d^m_E(W)$ generates $\mathcal{P}^m_{X/k} E$ as an $\mathcal{O}_X$-module if, and only if, $d^m_{E_K}(W \otimes_k K)$ generates $\mathcal{P}^m_{X_K/K} E_K$ as an $\mathcal{O}_{X_K}$-module.

Now $d^m_E(W)$ generates $\mathcal{P}^m_X E$ as an $\mathcal{O}_X$-module if, and only if, the evaluation map

$$\alpha : d^m_E(W) \otimes_k \mathcal{O}_X \to \mathcal{P}^m_{X/k} E$$

that sends $s \otimes g \mapsto gs$ is surjective. The first projection $u : X_K \to X$ is faithfully flat, so α is surjective if, and only if, its pullback

$$d^i_E(W) \otimes_k \mathcal{O}_{X_K} \xrightarrow{u^*\alpha} u^*\mathcal{P}^i_{X/k}E$$

is surjective.

The composition

$$E \longrightarrow u_*E_K \xrightarrow{u_*(d^i_{E_K})} u_*\mathcal{P}^i_{X_K/K}(E_K),$$

is a k-linear differential operator of order m, hence factors through an $\mathcal{O}_X$-linear map

$$\mathcal{P}^i_{X/k}E \longrightarrow u_*\mathcal{P}^i_{X_K/K}(E_K),$$

which by adjunction induces an $\mathcal{O}_{X_K}$-linear map

$$u^*\mathcal{P}^i_{X/k}E \xrightarrow{\beta} \mathcal{P}^i_{X_K/K}(E_K).$$

It is a standard fact that β is an isomorphism: sheaves of principal parts are stable base change. Therefore $u^*\alpha$ is surjective if, and only if, the composition $\beta \circ u^*\alpha$ is surjective. The result follows. □

3. The differential of a family of jets of sections

This section is the technical heart of this paper. The central notion is Defintion 3.6, which extends to families of jets the very simple construction of Remark 3.1. At the end of the section we prove a key lemma that is used in the proof of Theorem 0.1.

Let k be a field. Let X be a scheme over k. Let E be a locally free sheaf of finite rank on X. Let $V := \mathbf{V}(E)$ be the corresponding vector bundle. Let $\pi : V \to X$ be the projection.

{rmk:diff-single-jet}

Remark 3.1. Let $x \in X$ be a k-rational point. Let $s \in E/\mathfrak{m}_x^2E$ be a 1-jet of a section of E at x. Let $s(x) \in E/\mathfrak{m}_xE$ be the residue of s modulo $\mathfrak{m}_xE/\mathfrak{m}_x^2E$, which we identify with a k-rational point of V.

The 1-jet s has a natural differential, namely the k-linear map $ds : \mathrm{T}_X(x) \to \mathrm{T}_V(s(x))$ defined as follows: we choose a section germ $\tilde{s} \in E_x$ that reduces to s modulo $\mathfrak{m}_x^2E$, view $\tilde{s}$ as a germ at x of a morphism $X \to V$, and we set ds to be the differential of this morphism at x.

In the sequel we will need the following descriptions of the sheaf of differentials of a vector bundle and of the pullback map on 1-forms induced by a section of the corresponding locally free sheaf.

Let $h \in \Gamma(V, \pi^*E)$ be the tautological section, see Definition 1.1.

{vb-differentials}

Remark 3.2. The map $E^\vee \to \pi_*\Omega_{V/X}$ that sends $\tau \mapsto d(\tau \cdot h)$ is linear over $\mathcal{O}_X$. Its adjoint is an $\mathcal{O}_V$-linear isomorphism

$$\pi^*E^\vee \xrightarrow{\sim} \Omega_{V/X}$$

by Example 1.2 and the computation of the sheaf of differentials of affine space. It follows that the sheaf of differentials Ω_V is generated as an $\mathcal{O}_V$-module by its submodule $\pi^*\Omega_X$ and sections of the form $d(\tau \cdot h)$, where $\tau \in E^\vee$.

{rmk:diff-sect}

Remark 3.3. A section $\tilde{s} \in \Gamma(X, E)$ may be viewed as an element of $\mathrm{Hom}_X(X, V)$, hence induces an $\mathcal{O}_X$-linear pullback map on 1-forms

$$\tilde{s}^* : \tilde{s}^*\Omega_V \to \Omega_X,$$

which is uniquely characterized by the fact that $\tilde{s}^*(df) = df$ for all $f \in \mathcal{O}_X$, and $\tilde{s}^*d(\tau \cdot h) = d(\tau \cdot \tilde{s})$ for all $\tau \in E^\vee$. If X is smooth, the dual of $\tilde{s}^*$ is the differential $d\tilde{s} : \mathrm{T}_X \to \tilde{s}^*\mathrm{T}_V$ of $\tilde{s} : X \to V$.

Let $x : T \to X$ be a morphism of schemes. Let $s \in \Gamma(T, x^*\mathcal{P}^1_X E)$ be a section. Let $v \in \Gamma(T, x^*E)$ be the image of s under the truncation map $\mathcal{P}^1_X E \to E$. We identify v with a morphism of schemes $T \to V$ lifting x.

Motivated by the fact that the fiber of $\mathcal{P}^1_X E$ over a k-rational point $x \in X$ is canonically isomorphic to $E/\mathfrak{m}_x^2 E$, we think of s as a family of 1-jets of sections of E parametrized by T. Our next goal is to define the differential of s. This is done in Definition 3.6 below.

Remark 3.4. If $U \subseteq X$ be an open subset and $\tau \in \Gamma(U, E^\vee)$ is a section, then

$$d^1_{E^\vee}\tau \in \Gamma(U, \mathcal{P}^1_X \mathcal{H}om_{\mathcal{O}_X}(E, \mathcal{O}_X)) = \Gamma(U, \mathcal{H}om_{\mathcal{P}^1_X}(\mathcal{P}^1_X E, \mathcal{P}^1_X)),$$

so we have a natural product $(d^1_{E^\vee}\tau) \cdot s \in \Gamma(x^{-1}U, x^*\mathcal{P}^1_X)$.

The universal derivation $d : \mathcal{O}_X \to \Omega_X$ is a first-order differential operator. By the universal property of sheaves of principal parts, there exists a unique $\mathcal{O}_X$-linear map $\bar{d}$ such that the following diagram commutes:

$$\begin{tikzcd} \mathcal{O}_X \arrow[r, "d"] \arrow[dr, "d"'] & \mathcal{P}^1_X \arrow[d, dashed, "\bar{d}"] \\ & \Omega_X \end{tikzcd}$$

{diff-jet-prop}

Proposition 3.5. *There exists a unique map of $\mathcal{O}_T$-modules*

$$s^* : v^*\Omega_V \to x^*\Omega_X$$

such that $s^*(df) = df$ and $s^*(d(\tau \cdot h)) = \bar{d}(d^1_{E^\vee}\tau \cdot s)$ for all local sections $f \in \mathcal{O}_X$ and $\tau \in E^\vee$.

The proof of Proposition 3.5 will be given after Remark 3.7 below.

{diff-jet-def}

Definition 3.6. The $\mathcal{O}_T$-linear map $s^* : v^*\Omega_V \to x^*\Omega_X$ of Proposition 3.5 is called the *pullback map on 1-forms* induced by $s \in \Gamma(T, x^*\mathcal{P}^1_X E)$. If X is smooth over k, the dual $ds : x^*\mathrm{T}_X \to v^*\mathrm{T}_V$ of this $\mathcal{O}_T$-linear map is called the *differential* of s.

{rmk:diff-jet-funct}

Remark 3.7. (1) If $\tilde{s} \in \Gamma(X, E)$, then

$$d\tilde{s} = d(d^1_X \tilde{s})$$

as elements of $\mathrm{Hom}_X(\mathrm{T}_X, \tilde{s}^*\mathrm{T}_V)$. To see this, we compare Definition 3.5 with Remark 3.3 while noting that

$$\bar{d}(d^1_{E^\vee}\tau \cdot d^1_E s) = \bar{d}(d^1_X(\tau \cdot s)) = d(\tau \cdot s).$$

(2) If $a : T' \to T$ is a morphism of schemes, then

$$d(a^* s) = a^* ds$$

as elements of $\mathrm{Hom}_{T'}(a^* x^* \mathrm{T}_X, a^* v^* \mathrm{T}_V)$. This observation follows immediately from Definition 3.5.

Proof of Proposition 3.5. Uniqueness follows from Remark 3.2, so it suffices to show existence Zariski-locally on T, hence on X. Hence we may assume that E is trivial and choose an $\mathcal{O}_X$-linear basis $v_1, \ldots, v_e \in \Gamma(X, E)$.

Let $t_1, \ldots, t_r \in \Gamma(V, \mathcal{O}_V)$ be the unique sections such that

$$h = t_1 \cdot v_1 + \cdots + t_e \cdot v_e$$

as elements of $\Gamma(V, \pi^* E)$. Let $\mathbf{A}^e_k$ be the affine space of dimension e over k. There exists a unique isomorphism of X-schemes $V \cong X \times \mathbf{A}^e_k$ which identifies $t_1, \ldots, t_e$ with the coordinates on $\mathbf{A}^e_k$. Therefore

$$\Omega_V = \pi^* \Omega_X \oplus (\oplus_{l=1}^e \mathcal{O}_V \cdot dt_l).$$

The sheaf $\mathcal{P}^1_X E$ is a free $\mathcal{P}^1_X$-module with basis $d^1_E v_1, \ldots, d^1_E v_e$. Let $s_1, \ldots, s_e \in \Gamma(T, x^* \mathcal{P}^1_X)$ be the unique sections such that

$$s = s_1 \cdot d^1_E v_1 + \cdots + s_e \cdot d^1_E v_e$$

as elements of $\Gamma(T, x^* \mathcal{P}^1_X E)$. Let $s^* : v^* \Omega_V \to x^* \Omega_X$ be the unique $\mathcal{O}_T$-linear map that restricts to the identity on $x^* \Omega_X \subseteq v^* \Omega_V$ and sends $dt_l \mapsto \bar{d} s_l$ for all $l = 1, \ldots, e$.

Let $U \subseteq X$ be an open subset and let $\tau \in \Gamma(U, E^\vee)$ be a section. To complete the proof, let's show that

$$s^*(d(d^1_{E^\vee}\tau \cdot h)) = \bar{d}(\tau \cdot s)$$

as elements of $\Gamma(x^{-1}U, x^* \Omega_X)$.

Let $v_1^\vee, \ldots, v_e^\vee \in \Gamma(X, E^\vee)$ be the basis of $E^\vee$ dual to $v_1, \ldots, v_e \in \Gamma(X, E)$. Let $\tau_1, \ldots, \tau_e \in \Gamma(U, \mathcal{O}_X)$ be the unique sections such that

$$\tau = \tau_1 v_1^\vee + \cdots + \tau_e v_e^\vee \in \Gamma(U, E^\vee).$$

Then

$$\begin{aligned} d^1_{E^\vee}\tau \cdot s &= d^1_{E^\vee}\tau \cdot \textstyle\sum_l s_l d^1_E v_l \\ &= \textstyle\sum_l s_l (d^1_{E^\vee}\tau \cdot d^1_E v_l) \\ &= \textstyle\sum_l s_l d^1_X (\tau \cdot v_l) \\ &= \textstyle\sum_l s_l d^1_X \tau_l \end{aligned}$$

as elements of $\Gamma(x^{-1}U, x^*\mathcal{P}^1_X)$. Let $\bar{s}_1, \ldots, \bar{s}_e \in \Gamma(T, x^*E)$ be the images of $s_1, \ldots, s_e \in \Gamma(T, x^*\mathcal{P}^1_X E)$ under the truncation map $\mathcal{P}^1_X E \to E$. Then

$$v = \bar{s}_1 v_1 + \cdots + \bar{s}_e v_e$$

as elements of $\Gamma(T, x^*E) = \mathrm{Hom}_X(T, V)$. Furthermore,

$$\begin{aligned}
\bar{d}(d^1_{E^\vee}\tau \cdot s) &= \bar{d}\left(\textstyle\sum_l s_l d^1_X \tau_l\right) \\
&= \textstyle\sum_l \bar{d}s_l \cdot \tau_l + \bar{s}_l \cdot d\tau_l \\
&= \textstyle\sum_l s^* dt_l \cdot \tau_l + (v^\# t_l) \cdot d\tau_l \\
&= s^* d\left(\textstyle\sum_l t_l \tau_l\right) \\
&= s^* d(h \cdot \tau)
\end{aligned}$$

as elements of $\Gamma(x^{-1}U, x^*\Omega_X)$, where the second equality follows from the Leibniz rule proved below. The result follows. □

The following result was used in the proof of Proposition 3.5.

{prop:leibniz-pp}

Proposition 3.8 (Leibniz rule)**.** *Let $f, g \in \mathcal{P}^1_X$ be sections defined over a common open subset of X, and let $\bar{f}, \bar{g} \in \mathcal{O}_X$ be their respective images under the truncation map $\mathcal{P}^1_X \to \mathcal{O}_X$. Then*

$$\bar{d}(fg) = \bar{d}f \cdot \bar{g} + \bar{f} \cdot \bar{d}g \tag{3.1}$$

{eqn:leibniz}

as sections of Ω_X.

Proof. Both sides of (3.1) are $\mathcal{O}_X$-bilinear functions of f and g. As $\mathcal{P}^1_X$ is generated as an $\mathcal{O}_X$-module by the image of the universal differential operator $d^1_X : \mathcal{O}_X \to \mathcal{P}^1_X$, we may assume that $f = d^1_X u$ and $g = d^1_X v$ for some $u, v \in \mathcal{O}_X$. In this case, (3.1) reduces to the usual Leibniz rule satisfied by the universal derivation $d : \mathcal{O}_X \to \Omega_X$. □

Our next result is a lemma that will be used in the proof of the main result of this paper (Theorem 0.1). Let

$$0 \longrightarrow \Omega_X \otimes E \xrightarrow{\ \iota\ } \mathcal{P}^1_X E \longrightarrow E \longrightarrow 0,$$

be the canonical right-exact sequence. Let

$$\pi^*(\Omega_X \otimes E) = \mathcal{H}om_V(\pi^* E^\vee, \pi^*\Omega_X) \xrightarrow{\ \theta\ } \mathcal{H}om_V(\Omega_V, \pi^*\Omega_X)$$

be the $\mathcal{O}_V$-linear map induced by the canonical surjection $\Omega_V \twoheadrightarrow \Omega_{V/X} = \pi^* E^\vee$, see Remark 3.2.

{lemma:jet-vb-equiv}

Lemma 3.9. *Let $\delta \in \Gamma(T, x^*(\Omega_X \otimes E))$ be a section. Then the images of $s + \iota(\delta)$ and s under the truncation map $\mathcal{P}^1_X E \to E$ are equal, and the pullback maps*

$$s^*, (s + \iota(\delta))^* \in \mathrm{Hom}_T(v^*\Omega_V, x^*\Omega_X)$$

are related by the equation

$$(s + \iota(\delta))^* = s^* + \theta(\delta).$$

Proof. The difference

$$(s+\iota(\delta))^* - s^* : v^*\Omega_V \to x^*\Omega_X$$

sends $df \mapsto 0$ for all $f \in \mathcal{O}_X$, and sends

$$d(\tau \cdot h) \mapsto \bar{d}(d^1_{E^\vee}\tau \cdot \iota(\delta))$$

for all $\tau \in E^\vee$. The map $\theta(\delta)$, on the other hand, is equal to the composition

$$v^*\Omega_V \twoheadrightarrow v^*\Omega_{V/X} \cong v^*E^\vee \xrightarrow{\cdot\delta} x^*\Omega_X,$$

so sends $df \mapsto 0$ for all $f \in \mathcal{O}_X$, and

$$d(d^1_{E^\vee}\tau \cdot h) \mapsto \tau \cdot \delta$$

for all $\tau \in E^\vee$. To complete the proof, let us show that, given an open subset $U \subseteq X$ and a section $\tau \in \Gamma(U, E^\vee)$, we have

$$\bar{d}(\tau \cdot \iota(\delta)) = \tau \cdot \delta \tag{3.2}$$

as elements of $\Gamma(x^{-1}U, x^*\Omega_X)$.

Fix such U and τ. After shrinking X (and T), we may assume that $U = X$. Both sides of (3.2) are $\Gamma(T, \mathcal{O}_T)$-linear functions of δ, so after shrinking T, we may assume that there exist $f \in \Gamma(X, \mathcal{O}_X)$ and $w \in \Gamma(X, E)$ such that $\delta = df \otimes w$ as elements of $\Gamma(T, x^*(\Omega_X \otimes E))$. Now we may verify (3.2) directly:

$$\begin{aligned}
\bar{d}(d^1_E\tau \cdot \iota(\delta)) &= \bar{d}(d^1_E\tau \cdot (d^1_X f - f)d^1_E w) \\
&= \bar{d}(d^1_X(f\tau w) - f d^1_X(\tau w)) \\
&= d(f\tau w) - f d(\tau w) \\
&= df \cdot \tau w \\
&= \tau \cdot \delta
\end{aligned}$$

□

4. Proof of Theorem 0.1

Let k be a field. Let X be a smooth algebraic scheme of dimension n over k. Let E be a locally free $\mathcal{O}_X$-module of finite rank e. Let $V := \mathbf{V}(E)$ be the vector bundle associated to E. Let $\pi : V \to X$ be the projection. Let $Z \subseteq V$ be a locally closed subscheme. Suppose that Z is smooth over k and of pure codimension c in V.

{nd}

Definition 4.1. Let $s \in \Gamma(X, E)$ be a global section of E, which we identify with a section of the morphism $\pi : V \to X$. The *normal differential* of s with respect to Z, denoted $\mathbf{d}_Z s$, is the composition of the natural $\mathcal{O}_{s^{-1}Z}$-linear maps

$$\mathrm{T}_X|_{s^{-1}Z} \xrightarrow{ds} s^*(\mathrm{T}_V)_Z \twoheadrightarrow s^*(\mathrm{T}_V)_Z/s^*\mathrm{T}_Z = s^*\mathrm{N}_Z.$$

{transversality}

Remark 4.2. By definition, the morphism $s : X \to V$ is transverse to Z if, and only if, the following equivalent statements hold:

(1) The normal differential $\mathbf{d}_Z s$ is surjective.

(2) The scheme-theoretic inverse image $s^{-1}Z$ is smooth over k and of pure codimension c in X.

The equivalence of these statements follows from [EGA, Proposition IV.17.13.2].

The normal differential $\mathbf{d}_Z s$ can only be surjective if n (the rank of $\mathrm{T}_X|_{s^{-1}Z}$) is larger than or equal to c (the rank of $s^*\mathrm{N}_Z$). In this case, the first degeneracy locus (see Definition 1.4)

$$\Sigma^1(\mathbf{d}_Z f) \subseteq s^{-1}Z$$

is "bad locus" where s is not transverse to Z.

The construction of the normal differential can be extended to families of 1-jets of sections of E:

{def:nd-jet}

Definition 4.3. Let $x : T \to X$ be a morphism of schemes. Let $s \in \Gamma(T, x^*\mathcal{P}^1_X E)$ be a section, which we think of as a family of 1-jets of sections of E parametrized by T. Let $v \in \Gamma(T, x^*E)$ be the image of s under the truncation map $\mathcal{P}^1_X E \to E$. We identify v with a morphism of schemes $T \to V$ lifing x.

The *normal differential* of s with respect to Z, denoted $\mathbf{d}_Z s$, is the composition of the $\mathcal{O}_T$-linear maps

$$x^*\mathrm{T}_X|_{v^{-1}Z} \xrightarrow{ds} v^*\mathrm{T}_V|_{v^{-1}Z} \twoheadrightarrow v^*(\mathrm{T}_V|_Z)/v^*T_Z = v^*\mathrm{N}_Z,$$

where $ds : x^*\mathrm{T}_X \to v^*\mathrm{T}_V$ denotes the differential of s, see Definition 3.6.

We now consider the normal differential of the *universal* family of 1-jets of sections of E. Let $J := \mathbf{V}(\mathcal{P}^1_X E)$ be the vector bundle over X associated to the locally free $\mathcal{O}_X$-module $\mathcal{P}^1_X E$. Let $h \in \Gamma(J, (\mathcal{P}^1_X E)_J)$ be the tautological section, see Definition 1.1. Let $q \in \Gamma(J, E_J)$ be the image of h under the truncation map $\mathcal{P}^1_X E \to E$, which we identify with a morphism of X-schemes

$$q : J \to V.$$

The normal differential of h with respect to Z is an $\mathcal{O}_{q^{-1}Z}$-linear map

$$\mathbf{d}_Z h : (\mathrm{T}_X)_{q^{-1}Z} \to q^*\mathrm{N}_Z.$$

The first degeneracy locus $\Sigma^1(\mathbf{d}_Z h)$ of this map is a closed subscheme of $q^{-1}Z$, hence of J. Motivated by Remark 4.2 and the next proposition, we will refer to $\Sigma^1(\mathbf{d}_Z h)$ as the *universal bad locus* in J.

{prop:univ-sing-locus}

Proposition 4.4. *Let $s \in \Gamma(X, E)$ be a section. Then the scheme-theoretic inverse image along $d^1_E s \in \Gamma(X, \mathcal{P}^1_X E)$ of the universal bad locus $\Sigma^1(\mathbf{d}_Z h) \subseteq J$ is equal to $\Sigma^1(\mathbf{d}_Z s)$. In symbols:*

$$(d^1_E s)^{-1}\Sigma^1(\mathbf{d}_Z h) = \Sigma^1(\mathbf{d}_Z s)$$

Proof. The truncation map $\mathcal{P}^1_X E \to E$ sends $d^1_E s \mapsto s$, so $q \circ d^1_E s = s$ as morphisms from X to V, and therefore $(d^1_E s)^{-1} q^{-1} Z = s^{-1}Z$ as subschemes

of X. The normal differential $\mathbf{d}_Z h$ is a map of locally free modules on $q^{-1}Z$, so

$$\begin{aligned}(d^1_E s)^{-1}\Sigma^1(\mathbf{d}_Z h) &= (d^1_E s|_{s^{-1}Z})^{-1}\Sigma^1(\mathbf{d}_Z h)\\ &= \Sigma^1((d^1_E s|_{s^{-1}Z})^*\mathbf{d}_Z h),\end{aligned}$$

where the second equality follows from Remark 1.5 (4). We have the following equalities between elements of $\mathrm{Hom}_{s^{-1}Z}(\mathrm{T}_X|_{s^{-1}Z}, s^*\mathrm{T}_V|_Z)$:

$$\begin{aligned}(d^1_E s|_{s^{-1}Z})^* dh &= d((d^1_E s|_{s^{-1}Z})^* h)\\ &= d(d^1_E s|_{s^{-1}Z})\\ &= d(d^1_X s)|_{s^{-1}Z}\\ &= ds|_{s^{-1}Z}\end{aligned}$$

Indeed, the first and third equalities follow from Remark 3.7 (2), the second follows from the definition of the tautological section, and the fourth follows from 3.7 (1). Hence

$$(d^1_E s|_{s^{-1}Z})^*\mathbf{d}_Z h = \mathbf{d}_Z s,$$

which completes the proof. □

{prop:heart}

Proposition 4.5. *If $Z \subseteq V$ is smooth over X, then the universal bad locus $\Sigma^1(\mathbf{d}_Z h) \subseteq q^{-1}Z$ has pure codimension $|n-c|+c+1$ in J.*

Proof. Let H denote the vector bundle over Z corresponding to the locally free $\mathcal{O}_Z$-module $\mathcal{H}om_Z((\mathrm{T}_X)_Z, \mathrm{N}_Z)$. In symbols:

$$H := \mathbf{V}(\mathcal{H}om_Z((\mathrm{T}_X)_Z, \mathrm{N}_Z)).$$

Let $\alpha : (\mathrm{T}_X)_H \to (\mathrm{N}_Z)_H$ be the tautological $\mathcal{O}_H$-linear map. Consider the following diagram with Cartesian squares:

$$\begin{tikzcd}
\Sigma^1(\alpha) \arrow[d, hook] & \Sigma^1(\mathbf{d}_Z h) \arrow[l] \arrow[d, hook] & \\
H & q^{-1}Z \arrow[l, "\mathbf{d}_Z h"] \arrow[r, hook] \arrow[d] & J \arrow[d, "q"] \\
 & Z \arrow[r, hook] & V
\end{tikzcd}$$

The morphism $q : J \to V$ is smooth and surjective in view of the natural short exact sequence

$$0 \longrightarrow \Omega_X \otimes E \longrightarrow \mathcal{P}^1_X E \longrightarrow E \longrightarrow 0. \tag{4.1}$$

Therefore $q^{-1}Z$ has pure codimension c in J. The degeneracy locus $\Sigma^1(\alpha)$ is of pure codimension $|n-c|+1$ in H by Proposition 1.7. We will show that $\mathbf{d}_Z h : q^{-1}Z \to H$ is smooth and surjetive. This will imply that $\Sigma^1(\mathbf{d}_Z h)$ is of pure codimension $|n-c|+1$ in $q^{-1}Z$ and hence the result.

The short exact sequence (4.1) gives the V-scheme J the structrue of a principal bundle under the additive group V-scheme $\mathbf{V}((\Omega_X \otimes E)_V)$. This

structure induces on the Z-scheme $q^{-1}Z$ the structure of a principal bundle under the additive group Z-scheme $\mathbf{V}((\Omega_X \otimes E)_Z)$.

The composition of natural $\mathcal{O}_Z$-linear maps

$$E_Z = \mathrm{T}_{V/X}|_Z \hookrightarrow \mathrm{T}_V|_Z \twoheadrightarrow \mathrm{N}_Z$$

is surjective by the hypothesis that Z is smooth over X. This implies that the morphism of additive group Z-schemes

$$\Theta : \mathbf{V}((\Omega_X \otimes E)_Z) \to \mathbf{V}(\mathcal{H}om_Z((\mathrm{T}_X)_Z, \mathrm{N}_Z)) = H.$$

induced by it is smooth and surjective.

The morphism of Z-schemes $\mathbf{d}_Z h : q^{-1}Z \to H$ is equivariant under the group Z-scheme $\mathbf{V}((\Omega_X \otimes E)_Z)$ by Lemma 3.9. As Θ is smooth and surjective, this implies that $\mathbf{d}_Z h$ is smooth and surjective. □

{thm:main}

Theorem 4.6 (=Theorem 0.1)**.** *Suppose that the field k is infinite. Suppose that $Z \subseteq V$ is smooth over X. Let $W \subseteq \Gamma(X, E)$ be a finite-dimensional k-linear subspace that separates jets of order 1. Let $s \in W$ be a general element. Then the locally closed subscheme $s^{-1}Z \subseteq X$ is either empty or smooth over k of pure dimension $n - c$.*

Proof. Proposition 1.3 implies that $s^{-1}Z$ is either empty or of pure codimension c in X. If $c > n := \dim X$, then $s^{-1}Z$ is empty. Hence we may assume that $c \leq n$. Then the singular locus of $s^{-1}Z$ is equal to the degeneracy locus $\Sigma^1(\mathbf{d}_Z s)$ by Remark 4.2. Viewing $d^1_E s \in \Gamma(X, \mathcal{P}^1_X E)$ as a morphism from X to $J = \mathbf{V}(\mathcal{P}^1_X E)$, we have

$$\Sigma^1(\mathbf{d}_Z s) = (d^1_E s)^{-1} \Sigma^1(\mathbf{d}_Z h);$$

this is Proposition 4.4. By Proposition 2.1, the hypothesis that $W \subseteq \Gamma(X, E)$ separates jets of order 1 implies that $d^1_E(W) \subseteq \Gamma(X, \mathcal{P}^1_X)$ generates $\mathcal{P}^1_X E$ as an $\mathcal{O}_X$-module. Applying Proposition 1.3, we conclude that $\Sigma^1(\mathbf{d}_Z s)$ is either empty or of codimension in X equal to the codimension of the universal bad locus $\Sigma^1(\mathbf{d}_Z h)$ in J. By Proposition 4.5, the latter codimension is equal to $n + 1$, so $\Sigma^1(\mathbf{d}_Z s)$ is empty. □

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